

Predictors of Overall Survival in Advanced Lung Cancer

A Survival Analysis Using the Veterans' Administration Lung Cancer Trial

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Research Question and Objective

Research question

Which clinical and demographic factors are independently associated with overall survival in patients with advanced lung cancer?

Objective

To apply survival analysis methods to a randomised clinical trial dataset and estimate associations between patient characteristics and time-to-death using Kaplan–Meier estimation and Cox proportional hazards regression.

Primary outcome

- Overall survival (time from randomisation to death, in days)

Main predictors

- Treatment arm, age, Karnofsky performance score, histological cell type, time since diagnosis, prior therapy

Data Source and Study Design

Data source

- Veterans' Administration Lung Cancer Trial
- Kalbfleisch & Prentice (1980); available as `veteran` in the R `survival` package
- Randomised two-arm clinical trial in patients with advanced inoperable lung cancer

Study design

- Randomised controlled trial
- Standard chemotherapy vs. test chemotherapy
- Survival time measured in days from randomisation

Analytical approach

- Kaplan–Meier survival estimation and log-rank testing
- Cox proportional hazards regression (three nested models)
- Proportional hazards assumption testing and stratified model
- Model discrimination via concordance statistic (C-index)

Cohort Description

Characteristic	Value
Total participants	137
Deaths (events)	128 (93.4%)
Censored	9 (6.6%)
Median overall survival	80 days (95% CI: 52–105)
Mean survival	121.6 days
Median age	62 years
Median Karnofsky score	60
Treatment: Standard	69
Treatment: Test	68

The event rate was 93.4%, providing adequate power for survival modelling. The very short median survival of 80 days reflects the advanced and aggressive nature of the disease in this cohort.

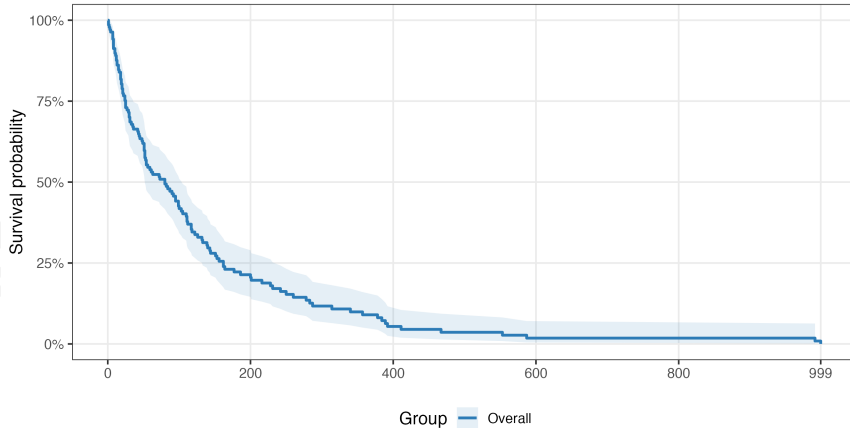
Descriptive Characteristics by Treatment Group

Characteristic	Standard (n = 69)	Test (n = 68)
Median survival, days	103 (95% CI: 59–132)	52.5 (95% CI: 44–95)
Median age, years	62	62
Median Karnofsky score	60	60
<i>Cell type, n (%)</i>		
Squamous	17 (25%)	18 (26%)
Small Cell	25 (36%)	23 (34%)
Adenocarcinoma	13 (19%)	14 (21%)
Large Cell	14 (20%)	13 (19%)
Log-rank p-value (treatment)	0.928	

The two treatment arms were well balanced at baseline. The unadjusted log-rank test showed no significant difference in survival between treatment arms ($p = 0.928$).

Overall Kaplan–Meier Survival Curve

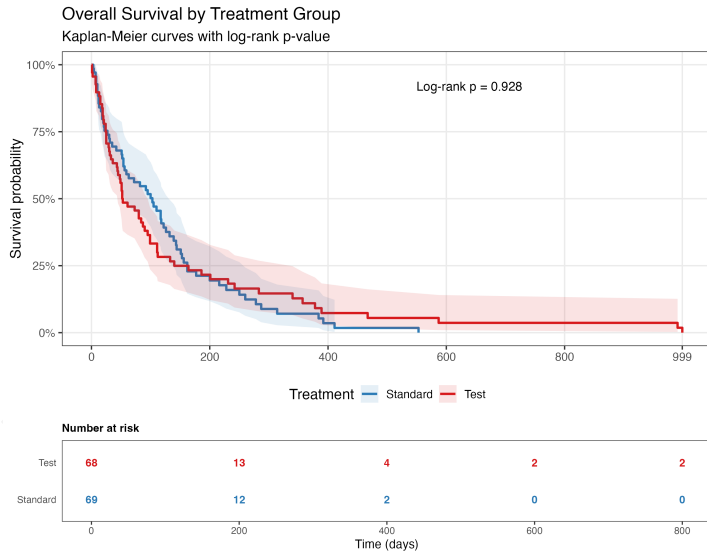
Overall Survival in Advanced Lung Cancer
Veterans' Administration Lung Cancer Trial (n = 137)



Number at risk



Survival by Treatment Group



Key results

- Standard arm median: 103 days (95% CI: 59–132)
- Test arm median: 52.5 days (95% CI: 44–95)
- Log-rank p = 0.928

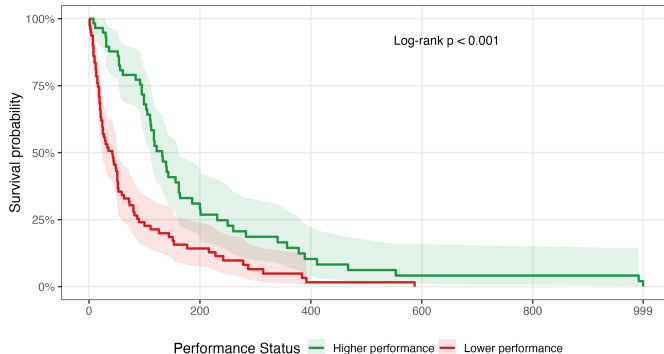
Interpretation

No statistically significant difference in survival between treatment arms. The curves overlap substantially throughout follow-up. This finding is confirmed in the adjusted Cox model (HR 1.34, p = 0.156).

Survival by Performance Status

Overall Survival by Performance Status

Karnofsky score grouped at threshold of 70



Number at risk

	0	200	400	600	800
Lower performance	79	10	1	0	0
Higher performance	58	15	5	2	2

Time (days)

Key results

- Higher performance median: 132 days (95% CI: 111–164)
- Lower performance median: 42 days (95% CI: 24–52)
- Log-rank $p < 0.001$

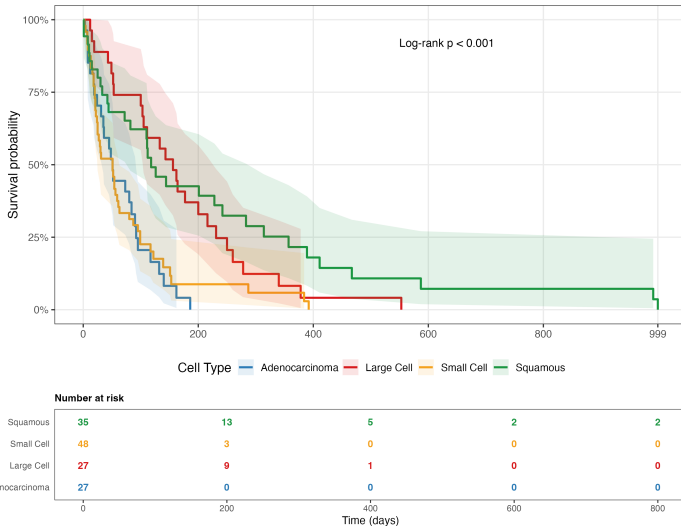
Interpretation

Performance status was the strongest unadjusted predictor. Patients with Karnofsky score ≥ 70 survived more than three times longer at the median compared to those with lower scores.

Survival by Histological Cell Type

Overall Survival by Histological Cell Type

Reference: Squamous cell carcinoma



Median survival

- Squamous: 118 days
- Large Cell: 156 days
- Small Cell: 51 days
- Adenocarcinoma: 51 days

Log-rank $p < 0.001$

Interpretation

Squamous and Large Cell carcinomas showed notably longer survival than Small Cell and Adenocarcinoma subtypes, consistent with established clinical evidence.

Log-Rank Test Summary

Comparison	χ^2	df	p-value
Treatment group	0.008	1	0.928
Performance status (Karnofsky ≥ 70 vs < 70)	20.26	1	<0.001
Cell type (4 groups)	25.40	3	<0.001
Prior therapy	0.501	1	0.479

Summary

- Performance status and cell type showed significant unadjusted associations with survival.
- Treatment group and prior therapy did not differ significantly in unadjusted analysis.
- These findings motivate the multivariable Cox regression to estimate adjusted associations.

Why Cox Proportional Hazards Regression?

Study objective

- Estimate independent predictors of survival time while adjusting for confounders.
- Account for censored observations — participants who did not die during follow-up.

Why Cox regression?

- Outcome is time-to-event (survival time), not binary — standard logistic regression is inappropriate.
- Cox model handles censoring correctly.
- Semi-parametric: no assumption required about the shape of the baseline hazard.
- Produces interpretable hazard ratios (HR) with 95% confidence intervals.
- Standard method in clinical survival analysis and oncology research.

Three nested Cox models were estimated to assess estimate stability:

Model	Covariates
Basic	Treatment, age
Clinical	Treatment, age, Karnofsky score, cell type
Full	Treatment, age, Karnofsky score, cell type, time since diagnosis, prior therapy

$$h(t) = h_0(t) \cdot \exp(\beta_1 \cdot \text{treatment} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{karno} + \dots)$$

Model evaluation

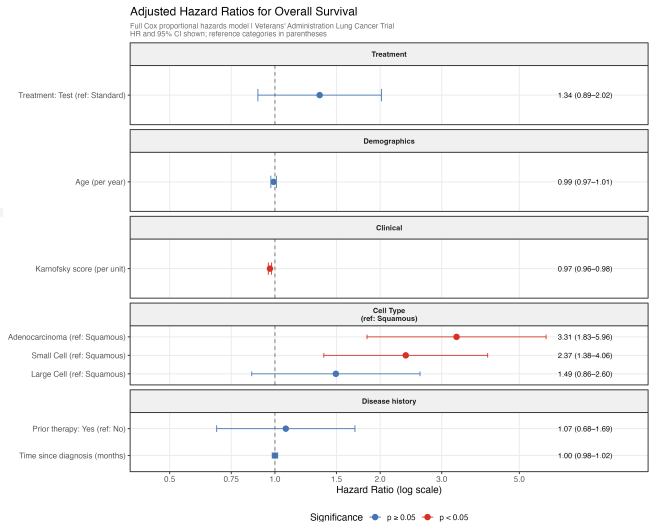
- Concordance statistic (C-index) for discrimination
- Schoenfeld residuals (`cox.zph`) for proportional hazards assumption

Main Cox Regression Results

Predictor	HR	95% CI	p-value
Treatment: Test vs Standard	1.34	0.89–2.02	0.156
Age (per year)	0.99	0.97–1.01	0.349
Karnofsky score (per unit)	0.97	0.96–0.98	<0.001
Small Cell vs Squamous	2.37	1.38–4.06	0.002
Adenocarcinoma vs Squamous	3.31	1.83–5.96	<0.001
Large Cell vs Squamous	1.49	0.86–2.60	0.156
Time since diagnosis	1.00	0.98–1.02	0.993
Prior therapy: Yes vs No	1.07	0.68–1.69	0.758

Hazard ratios adjusted for all variables in the full model. $HR > 1$ indicates higher hazard (worse survival); $HR < 1$ indicates lower hazard (better survival).

Adjusted Hazard Ratios — Forest Plot



Dashed line: $HR = 1$ (no effect). Red points: $p < 0.05$; blue points: $p \geq 0.05$. Error bars represent

Clinical Interpretation of Findings

Karnofsky performance score

- Each one-unit increase was associated with 3% lower hazard of death (HR 0.97, 95% CI 0.96–0.98, $p < 0.001$).
- The strongest and most consistent independent predictor across all model specifications.

Cell type

- Small Cell carcinoma: 2.37-fold higher hazard vs Squamous (95% CI 1.38–4.06, $p = 0.002$).
- Adenocarcinoma: 3.31-fold higher hazard vs Squamous (95% CI 1.83–5.96, $p < 0.001$).
- Large Cell: elevated point estimate (HR 1.49) but not statistically significant ($p = 0.156$).

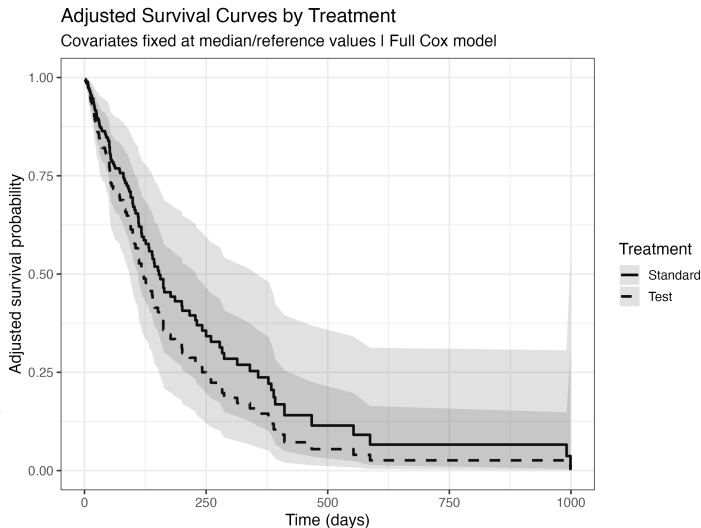
Treatment

- No significant survival advantage of the test treatment over standard (HR 1.34, $p = 0.156$).
- Finding is consistent across all three model specifications and confirmed by log-rank test.

Age, time since diagnosis, prior therapy

- None were independently associated with survival after full adjustment.

Adjusted Survival Curves by Treatment



Method

Predicted survival curves from the full Cox model with all covariates fixed at median or reference values, isolating the treatment contrast after adjustment.

Interpretation

The two curves are very close throughout follow-up, with overlapping confidence intervals. This confirms the Cox model result: no meaningful survival difference between treatment arms after covariate adjustment.

Sensitivity Analysis — Model Specification

Predictor	Basic HR	Clinical HR	Full HR
Treatment: Test vs Standard	1.00	1.35	1.34
Age (per year)	1.01	0.99	0.99
Karnofsky score	—	0.97	0.97

Concordance (C-index)

Model	C-index
Basic (treatment + age only)	0.514
Clinical (+ Karnofsky + cell type)	0.738
Full (all predictors)	0.736

Adding Karnofsky score and cell type accounted for almost all of the model's discriminative ability. The full model ($C = 0.736$) offers acceptable discrimination.

Proportional Hazards Assumption

Test: Schoenfeld residuals (`cox.zph`)

Variable	χ^2	p-value	Assumption met?
Treatment	0.264	0.607	Yes
Age	1.829	0.176	Yes
Karnofsky	12.935	<0.001	No — violation
Cell type	15.227	0.002	No — violation
Diagtime	0.013	0.910	Yes
Prior therapy	2.166	0.141	Yes
GLOBAL	34.552	<0.001	No — violation

Interpretation

The PH assumption was violated for Karnofsky score and cell type, meaning their effects on the hazard were not constant over time. A stratified Cox model was fitted as a remedy.

Stratified Cox Model — Remedy for PH Violation

Approach

Karnofsky performance status and cell type were moved from the hazard ratio estimation into the stratification structure. This allows a separate baseline hazard for each stratum, relaxing the PH assumption for those variables while preserving it for the remaining predictors.

Stratified model results

Predictor	HR	95% CI	p-value
Treatment: Test vs Standard	1.09	0.71–1.67	0.701
Age (per year)	1.00	0.98–1.02	0.959
Time since diagnosis	1.00	0.98–1.02	0.974
Prior therapy: Yes vs No	1.11	0.68–1.79	0.676

PH test on stratified model: GLOBAL $p = 0.153$ — assumption satisfied.

The treatment finding is unchanged: no significant survival advantage of the test treatment (HR 1.09, $p = 0.701$).

Conclusions

Main findings

- Median overall survival was 80 days (95% CI: 52–105), reflecting aggressive advanced disease.
- Karnofsky performance score was the strongest independent predictor of survival (HR 0.97 per unit, $p < 0.001$).
- Adenocarcinoma (HR 3.31) and Small Cell (HR 2.37) carcinomas were associated with substantially higher hazard of death compared to Squamous cell carcinoma.
- No significant survival advantage was found for the test treatment vs. standard in either unadjusted (log-rank $p = 0.928$) or adjusted analyses (HR 1.34, $p = 0.156$).
- Age, time since diagnosis, and prior therapy were not independently associated with survival.
- The full model demonstrated acceptable discrimination (C-index = 0.736).

Overall conclusion

Performance status and histological cell type are the dominant prognostic factors in advanced lung cancer. The test chemotherapy regimen did not provide a survival benefit over standard

Limitations

- Small sample size ($n = 137$); wide confidence intervals for several hazard ratio estimates, particularly for cell type subgroups.
- Very high event rate (93.4%) with only 9 censored observations, limiting the ability to assess long-term survival.
- The proportional hazards assumption was violated for Karnofsky score and cell type; a stratified model was applied but time-varying coefficient models could provide additional insight.
- The dataset is from a historical trial (circa 1980); treatment regimens do not reflect current oncological practice and findings should not be extrapolated to modern lung cancer management.
- No information available on disease stage, molecular markers, or comorbidities, which are important prognostic factors in contemporary lung cancer research.
- Results should be interpreted as an applied statistical portfolio analysis rather than a definitive clinical conclusion.

Software and packages

- R
- `survival` — core survival methods
- `tidyverse` — data wrangling
- `gtsummary` — summary tables
- `broom` — tidy model output
- `cowplot` — figure layout
- `ggplot2` — visualisation

Statistical methods

- Kaplan–Meier survival estimation
- Log-rank test (4 comparisons)
- Univariable and multivariable Cox regression
- Model specification sensitivity analysis
- Proportional hazards assumption (`cox.zph`)
- Stratified Cox model

Model evaluation

- Concordance statistic (C-index)
- Schoenfeld residual plots

Thank You

Questions?

Appendix: Full Cox Model Results

Predictor	HR	95% CI	p-value
Treatment: Test vs Standard	1.34	0.89–2.02	0.156
Age (per year)	0.99	0.97–1.01	0.349
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Time since diagnosis (months)	1.00	0.98–1.02	0.993
Prior therapy: Yes vs No	1.07	0.68–1.69	0.758
<i>Model diagnostics</i>			
Concordance (C-index)	0.736		
PH assumption (GLOBAL)	p < 0.001 — violated for karno, cell type		